Metals Market Index MMI

MMi Dashboard

Iron Ore Price Indices

Iron Ore Port Stock (FOT Qingdao)
IOPI62 62% Fe Fines RMB/t

928
16 1.75%
Mar 1st, 2023

Iron Ore Port Stock (FOT Qingdao)
IOPI65 65% Fe Fines RMB/t

1040
17.00 1.66%
Mar 1st, 2023

Iron Port Stock (FOT Qingdao)
IOPI58 58% Fe Fines RMB/t

831
7 0.85%
Mar 1st, 2023

Iron Ore Seaborne (CFR Qingdao)
IOSI62 62% Fe Fines USD/dmt

128.45
0.00 0.00%
Mar 1st, 2023

Iron Ore Seaborne (CFR Qingdao)
IOSI65 65% Fe Fines USD/dmt

143.20
0.00 0.00%
Mar 1st, 2023

Iron Ore Port Stock (FOT Qingdao)
IOPLI 62.5% Fe Lump RMB/t

1010
20 2.02%
Week Ending Feb 24th, 2023

Exchange Traded Contracts

DCE Iron Ore 62% Fines
I2305 (Jan) RMB/t (3pm close)

908.50
20.00 2.25%
Mar 1st, 2023 (3pm close)

SGX Iron Ore (CFR Qingdao) 62% Fe Fines
March 23 USD/dmt

126.25
2.85 2.31%
Mar 1st, 2023 (5.30 pm Print)

SHFE Rebar
RB2305 (Jan) RMB/t

4216
42 1.01%
Mar 1st, 2023 (3pm close)

Freight Rates

C3, Tubarao - Qingdao USD/t

17.14
0.02 0.13%
Feb 28th, 2023

C5, W. Australia - Qingdao USD/t

7.07
0.00 0.04%
Feb 28th, 2023

Steel Price

Steel Rebar (China Domestic) RMB/t

4260
80 1.91%
Week Ending Feb 24th, 2023

Inventory Levels

Iron Ore Inventory at Chinese Ports (35)
million tonnes

138.11
1.00 0.73%
Week Ending Feb 24th, 2023

Steel Inventory in China
million tonnes

19.48
-0.40 -1.99%
Week Ending Feb 24th, 2023

Steel Price

Steel HRC (China Domestic) RMB/t

4300
60 1.42%
Week Ending Feb 24th, 2023

IRON ORE PORT STOCK INDEX (IOPI)

Mar 1st, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOPI62	62% Fe Fines	928	16	1.8%	869	880	858	892	124.74	2.43	2.0%	120.06	120.58	117.93	122.93
IOPI58	58% Fe Fines	831	7	0.8%	813	798	761	793	112.52	1.17	1.1%	113.29	110.17	105.29	110.04
IOPI65	65% Fe Fines	1040	17	1.7%	981	992	970	1003	140.27	2.59	1.9%	136.05	136.43	133.84	138.73

IRON ORE SEABORNE INDEX (IOSI)

Mar 1st, 2023		CFR Qingdao, USD/dry tonne						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOSI62	62% Fe Fines	128.45	0.00	0.00%	108.32	122.48	83.90	159.45
IOSI65	65% Fe Fines	143.20	0.00	0.00%	118.76	139.49	94.45	185.82

MARKET COMMENTARY

DCE iron ore futures open lower and go higher, the main contract I2305 closed 908.5, an increase of 2.42% throughout the day. some traders were active to sell, but steel mills tended to be wait-and-see, and purchased on demand. today's overall market transaction atmosphere in general. PBF at Shandong port dealt 905-915 yuan/mt; increased 10-12 yuan/mt over yesterday. PBF at Tangshan port dealt 922-930 yuan/mt; increased 15-25 yuan/mt over yesterday. The apparent demand for steel in recent years has been average, but the PMI data of China's manufacturing industry in February released by the Bureau of Statistics today exceeded expectations, boosting market sentiment and driving up iron ore prices. In addition, the weather in Tangshan and Qinhuangdao of Hebei Province has improved, the sintering production restriction has ended, and the demand for iron ore has increased slightly, which may continue to support the strong and volatile operation of iron ore prices.

IRON ORE PORT LUMP INDEX (IOPLI)

Week Ending Feb 24th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [3]						
Index	Fe Content	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]
IOPLI62	62.5% Fe Lump	1010	20	2.0%	903	1035	770	1405	131.53	1.79	1.38%	116.26	138.76	94.72	199.13

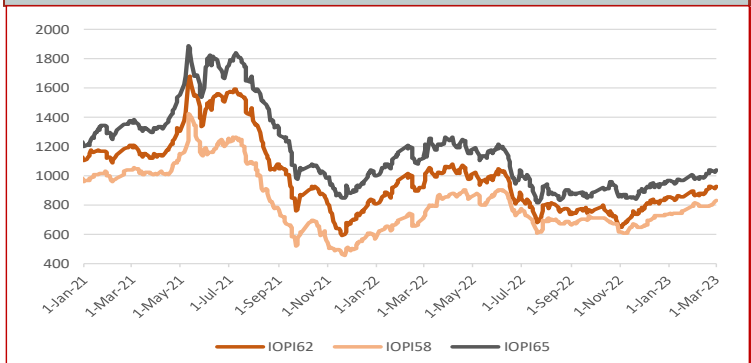
IRON ORE DOMESTIC CONCENTRATE SPOT PRICE ASSESSMENTS AND COMPOSITE INDEX

Week Ending Feb 24th, 2023				RMB/tonne (excluding tax) [3]				USD/tonne (excluding tax) [3]			
Province	Region	Product	Basis	This week	Change %	Low [2]	High [2]	This week	Change %	Low [2]	High [2]
Hebei	Hanxing	66% Fe Concentrate	Dry	1043	0.7%	779	1645	151.63	0.01%	110.31	255.69
Hebei	Qian'an	65% Fe Concentrate	Dry	1075	1.9%	780	1630	156.28	1.22%	110.51	251.57
Liaoning	Anshan	65% Fe Concentrate	Wet	810	2.5%	620	1310	117.76	1.85%	87.40	202.32
Shandong	Zibo	65% Fe Concentrate	Dry	1066	0.7%	800	1752	154.97	-0.01%	117.19	272.32

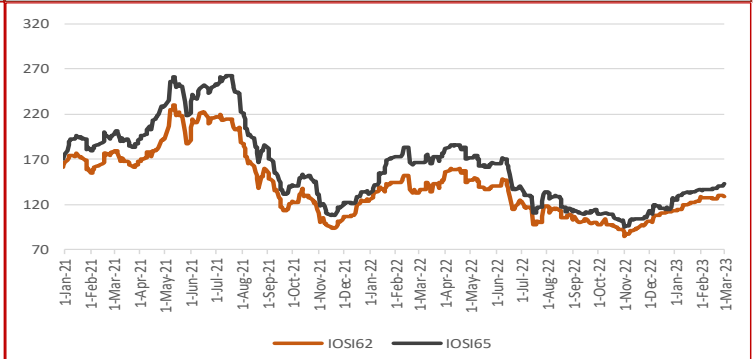
Week Ending Feb 24th, 2023	This week	Change %	Low [2]	High [2]
China Mines Concentrate Composite Index RMB/WT	949.64	1.27%	706.36	1511.22

[1] Exchange rate applied: RMB/USD = 6.7522 [2] Last 12 months
[3] Weekly exchange rate applied: RMB/USD =6.7849

IRON ORE PORT INDEX, FOT QINGDAO (RMB/WT)



IRON ORE SEABORNE INDEX, CFR QINGDAO (USD/DMT)



IRON ORE PORT STOCK INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Mar 1st, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne						
Index	Fe Content	November	December	January	February	MTD	QTD	YTD	November	December	January	February	MTD	QTD	YTD
IOPI62	62% Fe Fines	714	822	856	892	869	881	880	92.18	109.43	117.00	121.64	120.06	120.90	120.58
IOPI58	58% Fe Fines	635	711	751	799	813	792	798	82.39	95.17	103.34	109.79	113.29	109.35	110.17
IOPI65	65% Fe Fines	865	935	968	1003	981	993	992	112.37	124.90	132.80	137.36	136.05	136.72	136.43

IRON ORE SEABORNE INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Mar 1st, 2023		CFR Qingdao, USD/dry tonne						
Index	Fe Content	November	December	January	February	MTD	QTD	YTD
IOSI62	62% Fe Fines	93.05	108.92	119.76	127.73	108.32	108.53	122.48
IOSI65	65% Fe Fines	102.40	116.83	132.39	138.06	118.76	119.46	139.49

FREIGHT RATES

Feb 28th, 2023		FREIGHT RATES - DRY BULK US$/wet tonne				
Route	Designation		Change	Change %	Low [2]	High [2]
W. Australia - Qingdao	C5	7.07	0.00	0.04%	3.57	16.77
Tubarao - Qingdao	C3	17.14	0.02	0.13%	6.70	36.40

IRON ORE PORT LUMP INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Week Ending Feb 24th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	November	December	January	February	MTD	QTD	YTD	November	December	January	February	MTD	QTD	YTD
IOPLI62	62.5% Fe Lump	811	922	964	988	903	904	1035	100.83	117.06	126.47	129.95	116.26	116.88	138.76

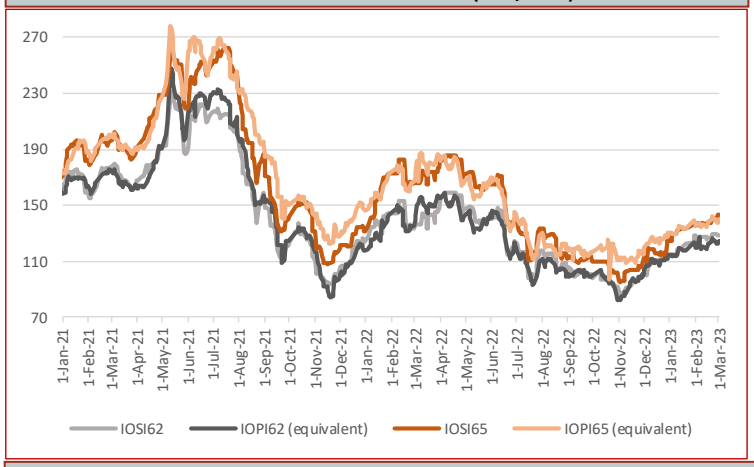
IRON ORE INDEX PREMIUMS/DISCOUNTS

Mar 1st, 2023		PORT STOCK INDEX (RMB/WT)	
Index	Fe Content	Spread to IOPI62	% Spread to IOPI62
IOPI58	58% Fe Fines	-97	-10.45%
IOPI65	65% Fe Fines	112	12.07%

Mar 1st, 2023		SEABORNE INDEX (USD/DMT)	
Index	Fe Content	Spread to IOSI62	% Spread to IOSI62
IOSI65	65% Fe Fines	14.75	11.48%

COPYRIGHT METALS MARKET INDEX, ALL RIGHTS RESERVED

IRON ORE INDEX COMPARISONS (USD/DMT)



CHINA DOMESTIC COMPOSITE MINES INDEX (RMB/Wet Tonne, including VAT)

IRON ORE BRAND SPOT PRICE ASSESMENTS

Mar 1st, 2023	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI62
Roy Hill	871	16	-57
SIMEC Fines	802	17	-126
PB Fines	895	15	-33
Newman Fines	897	16	-31
MAC Fines	878	16	-50
Jimblebar Blended Fines	793	16	-135
Carajas Fines	1109	16	181
Brazilian SSF	893	16	-35
Brazilian Blend Fines	909	16	-19
RTX Fines	811	16	-117
West Pilbara Fines	839	16	-89

Mar 1st, 2023	SEABORNE INDEX (USD/DMT)		
	Price	Change	Diff to IOSI62
Roy Hill	123.95	0.00	-4.50
SIMEC Fines	120.45	0.00	-8.00
PB Fines	124.70	0.00	-3.75
Newman Fines	127.60	0.00	-0.85
MAC Fines	124.70	0.00	-3.75
Jimblebar Blended Fines	117.10	0.00	-11.35
Carajas Fines	158.00	0.00	29.55
Brazilian SSF	132.20	0.00	3.75
Brazilian Blend Fines	133.85	0.00	5.40
RTX Fines	118.35	0.00	-10.10
West Pilbara Fines	122.70	0.00	-5.75

Mar 1st, 2023	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI58
SSF	831	7	0
FMG Blended Fines	841	7	10
Robe River	842	7	11
Western Fines	845	7	14
Atlas Fines	839	8	8
Yandi	823	7	-8

IRON ORE INDEX NORMALISATION DIFFERENTIALS

Port Stock Index Product Differentials (RMB/wet tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	5.00	0.00
	High Grade Fe 63 - 64%	5.00	0.00
	High Grade Fe 64 - 65%	5.00	0.00
	High Grade Fe 65 - 65.5%	5.00	0.00
	Low Grade Fe	41.00	3.00
1% Alumina	High Fe Grade Al <2.25%	84.00	-4.00
	High Fe Grade Al 2.25-4%	5.00	0.00
	Low Fe Grade Al <2.25%	25.00	-5.00
	Low Fe Grade Al 2.25-4%	15.00	0.00
1% Silica	High Fe Grade Si <4%	26.00	0.00
	High Fe Grade Si 4-6.5%	5.00	0.00
	Low Fe Grade	49.00	6.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	10.00	1.00
	High Fe Grade 0.115%<P<0.15%	10.00	1.00
	Low Fe Grade 0.09<P<0.1%	5.00	0.00

Seaborne Index Product Differentials (USD/dry tonne)

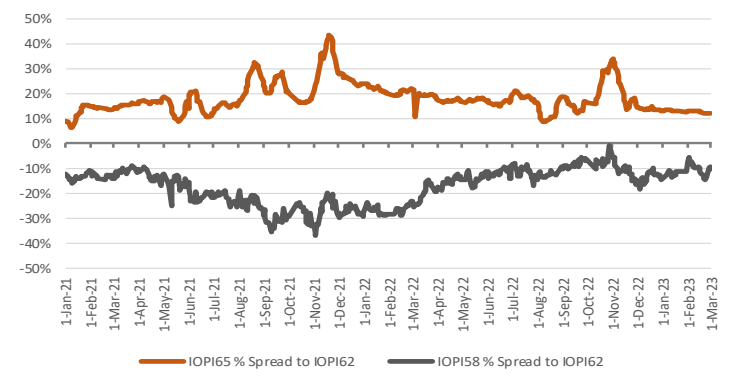
	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	3.00	0.25
	High Grade Fe 63 - 64%	4.00	0.00
	High Grade Fe 64 - 65%	4.00	0.00
	High Grade Fe 65 - 65.5%	4.00	0.00
1% Alumina	High Fe Grade Al <2.25%	3.50	0.50
	High Fe Grade Al 2.25-4%	3.25	0.50
1% Silica	High Fe Grade Si <4%	3.50	0.50
	High Fe Grade Si 4 - 6.5%	0.50	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	2.75	0.00
	High Fe Grade 0.115%<P<0.15%	4.75	0.00

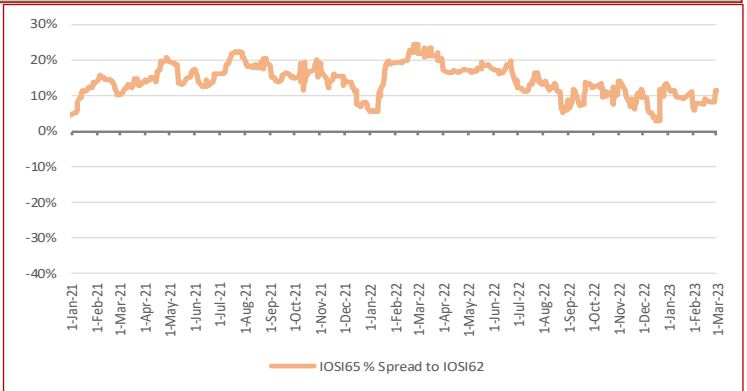
Port Stock Price Differentials to Qingdao Port for PB Fines (RMB/wet tonne)

Port	Value	Change	Port	Value	Change	Port	Value	Change	Port	Value	Change
Bayuquan	-100.00	0.00	Fangcheng	-25.00	0.00	Lanshan	0.00	0.00	Rizhao	0.00	0.00
Beilun	0.00	0.00	Jiangyin	-30.00	0.00	Lianyungang	0.00	0.00	Shekou	0.00	0.00
Caofeidian	-5.00	0.00	Jingtang	-5.00	0.00	Majishan	0.00	0.00	Taicang	-30.00	0.00
Dalian	0.00	0.00	Lanqiao	0.00	0.00	Qingdao	0.00	0.00	Tianjin	0.00	0.00

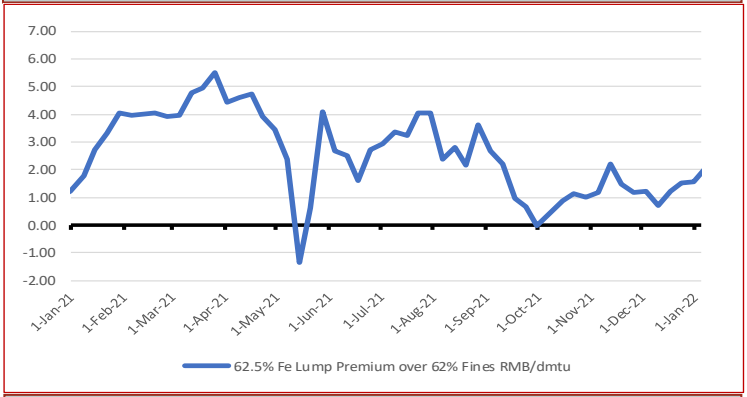
COPYRIGHT METALS MARKET INDEX, ALL RIGHTS RESERVED

IRON ORE INDEX PREMIUMS/DISCOUNTS

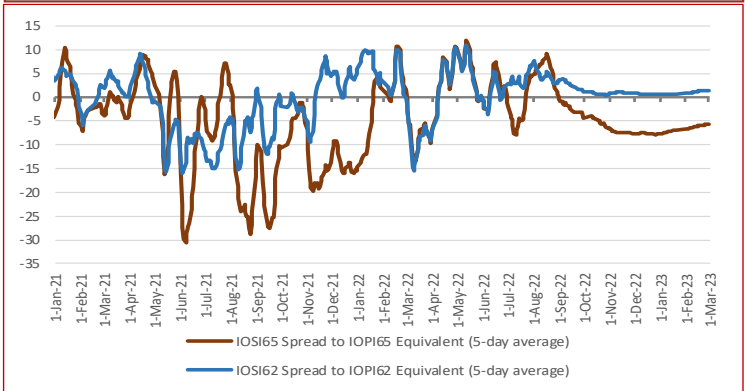




WEEKLY IRON ORE PORT STOCK LUMP PREMIUM (RMB/DMTU)

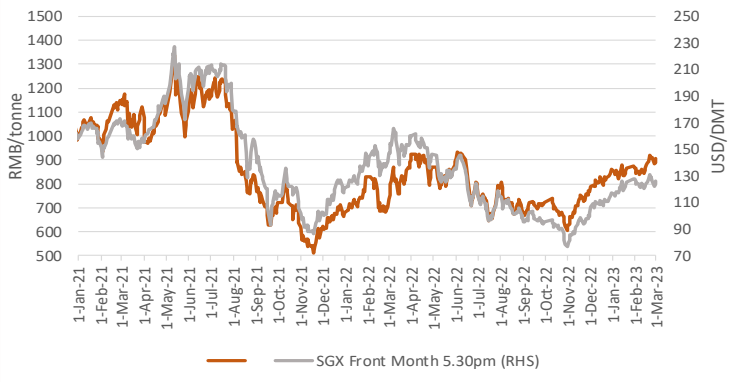


IRON ORE SEABORNE TO PORT STOCK PRICE SPREADS (USD/DMT)



TOTAL IRON ORE INVENTORIES AT CHINA PORTS (MILLION TONNES)

FUTURE TRADING—FRONT MONTH CLOSING PRICE



IRON ORE PORT INVENTORIES (MILLION TONNES)

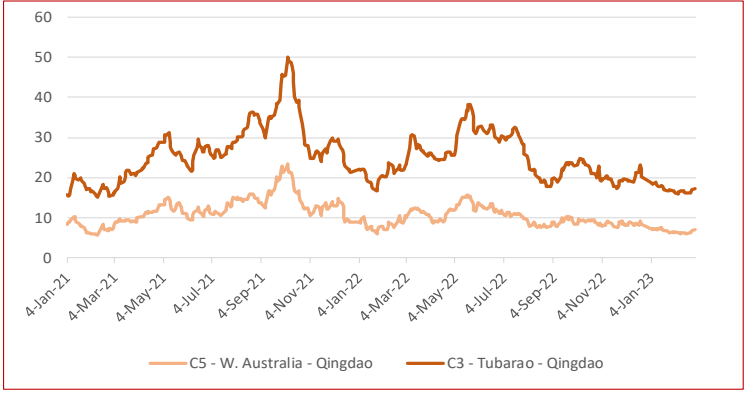
Week Ending Feb 24th, 2023

Province	This week	Change %	Low [2]	High [2]
Jingtang	12.04	-0.82%	9.84	19.20
Qingdao	25.94	-1.14%	9.41	26.24
Caofeidian	11.95	2.75%	9.05	16.29
Tianjin	12.13	2.10%	7.14	12.97
Rizhao	19.02	-0.68%	9.44	19.26
Total (35 Ports)	138.11	0.73%	98.80	155.39

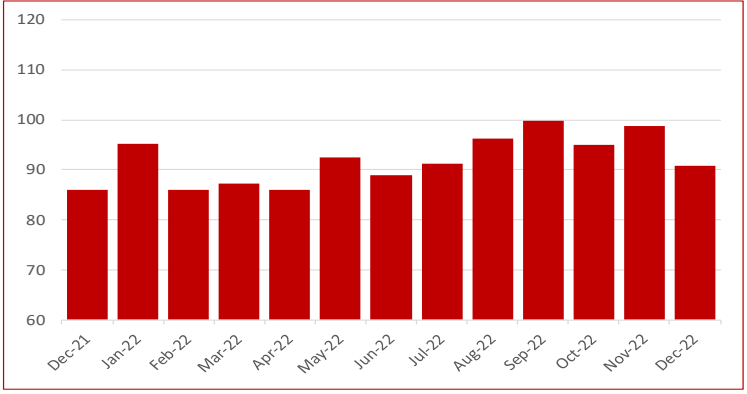
IRON ORE FUTURES CONTRACTS

	DCE (RMB/WMT)			SGX (USD/DMT)		
Closing Date	Mar 1st, 3pm close			Mar 1st, 5:30pm		
Contract	I2305	Change	Change %	Mar. 23	Change	Change %
Closing Price	908.50	20.00	2.25%	126.25	2.85	2.31%
Vol traded ('000 lots)	53.48	6.85	14.69%	3.67	-1.58	-30.16%
Open positions ('000 lots)	84.69	1.72	2.07%	27.49	-1.49	-5.14%
Day Low	885.5	2.50	0.28%	123.50	0.75	0.61%
Day High	910.5	18.50	2.07%	126.95	2.50	2.01%

DRY BULK FREIGHT RATES (USD/MT)



TOTAL CHINA IRON ORE IMPORT VOLUMES (MILLION TONNES)



COPYRIGHT METALS MARKET INDEX, ALL RIGHTS RESERVED

STEEL SPOT MARKET PRICES—CHINA

Steel Spot Market RMB/tonne

Product	2023/2/24	Change	Change %
ReBar HRB400 ϕ18mm	4,260	80	1.91%
Wirerod Q300 ϕ6.5mm	4,510	80	1.81%
HRC Q235/SS400 5.5mm*1500*C	4,300	60	1.42%
CRC SPCC/ST12 1.0mm*1250*2500	4,730	110	2.38%
Medium & Heavy Plate Q235B 20mm	4,370	70	1.63%
GI ST02Z 1.0mm*1000*C	4,960	90	1.85%
Colour Coated Plate	7,550	-30	-0.40%

CHINESE STEEL MILL PROFITABILITY

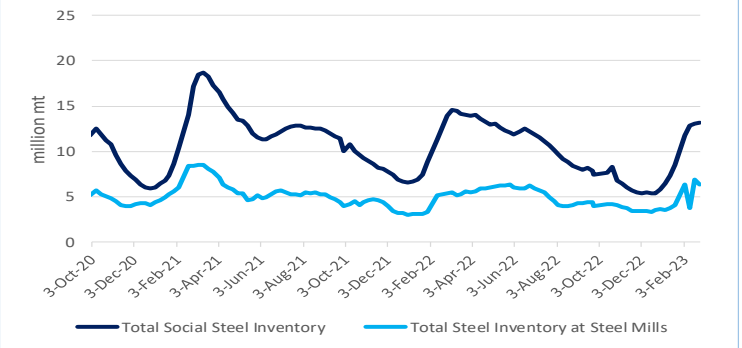
SMM Tracking of Steel Mill P&L - Rebar and Hot-rolled Coil (RMB/tonne)

Category	Price	Change (WoW)	Note
MMi (Fe 62%), USD/mt exluding tax	129.65	3	Mmi CFR Equivalent index for 1st Feb
Coke	2,960	0	2nd grade met coke, Tangshan, incl. tax
Steel Scrap	3,200	50	steel scrap (6mm) in Zhangjiagang, exl. tax
Billet Cost	3,677	64	Q234, incl. tax
Rebar cost - Blast furnace	4,189	62	calculated based on theoretical weight, incl. tax
Rebar profit - Blast furnace	-19	-82	based on Shanghai prices, incl. tax
Hot-rolled coil cost - Blast furnace	4,190	61	based on actual weight, incl. tax
Hot-rolled coil proft - Blast furnace	100	-11	based on Shanghai prices, incl. tax

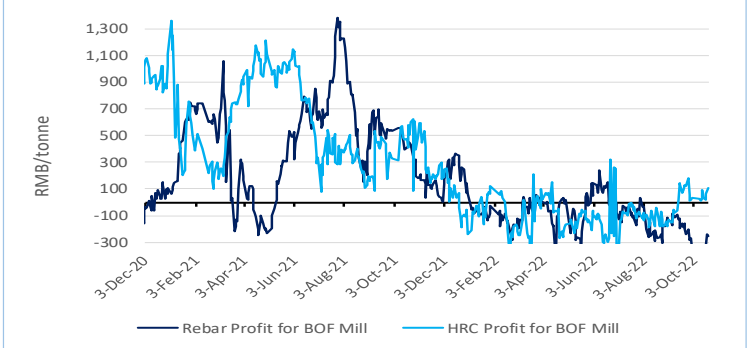
Note: 1. Costs in the table are caluclated based on todays market prices and facout our management, sales, financial and depreciations fe

2. The cost refers to average cost in the industry based on SMM's survey of small, medium and large mills in China

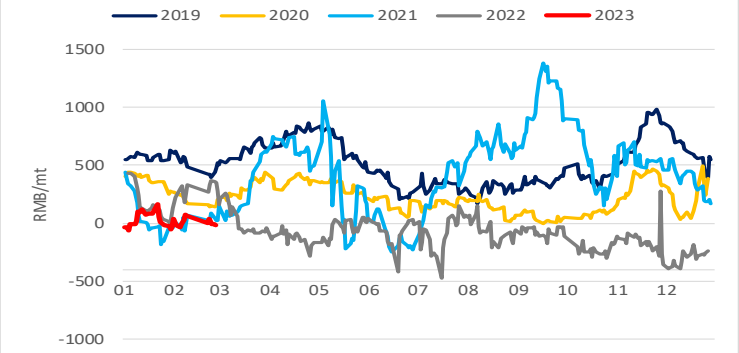
CHINESE STEEL INVENTORIES



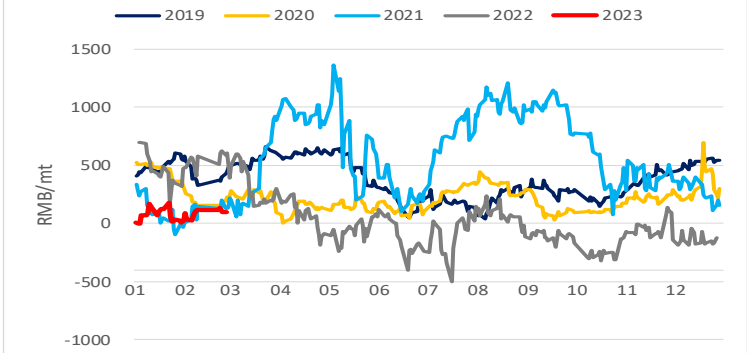
CHINESE STEEL MILL PROFITABILITY



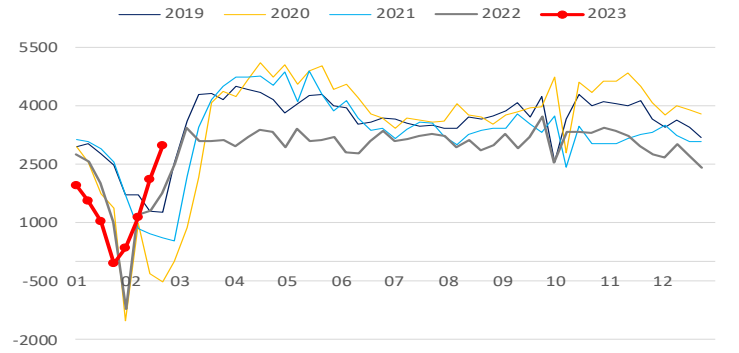
CHINESE STEEL MILL PROFITABILITY—Rebar



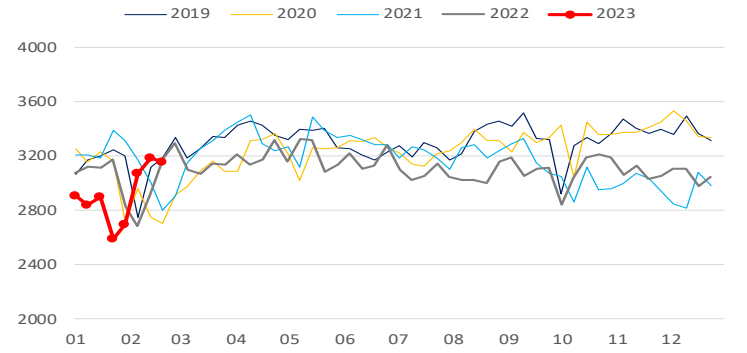
CHINESE STEEL MILL PROFITABILITY—Hot-rolled Coil



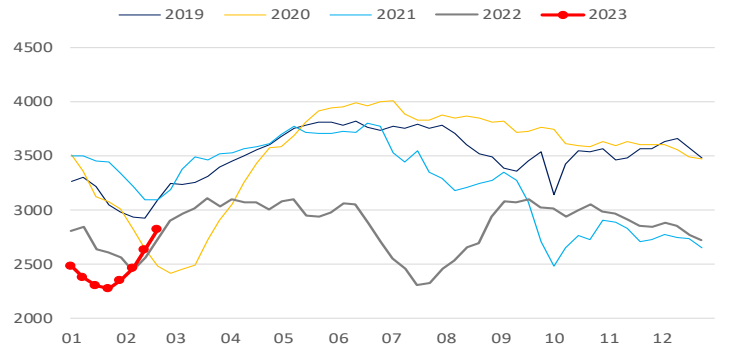
CHINESE STEEL CONSUMPTION—Rebar



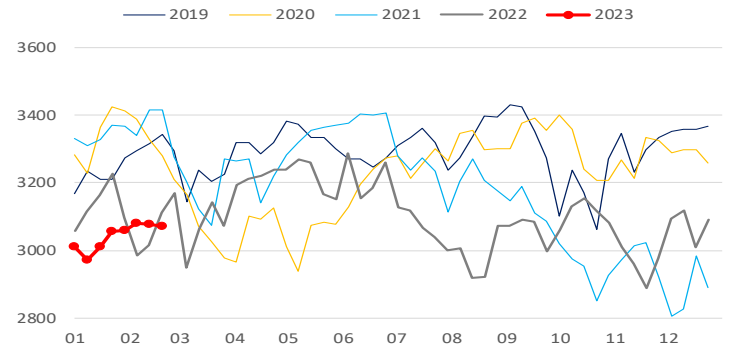
CHINESE STEEL CONSUMPTION—Hot-rolled Coil



CHINESE STEEL PRODUCTION—Rebar



CHINESE STEEL PRODUCTION—Hot-rolled Coil



COPYRIGHT METALS MARKET INDEX, ALL RIGHTS RESERVED

IRON ORE INDEX SPECIFICATIONS, COMPILATION RATIONALE AND DATA EXCLUSIONS

Iron Ore Index Specifications (Port and Seaborne)

	65% Fe Fines	62% Fe Fines	58% Fe Fines	62.5% Fe Lump
Fe %	65.00	62.00	58.00	62.50
Alumina %	1.40	2.25	2.25	1.50
Silica%	1.50	4.00	5.50	3.50
Phosphorus %	0.06	0.09	0.05	0.08
Sulphur %	0.01	0.02	0.02	0.02
Moisture %	8.00	8.00	9.00	4.00
Sizing	Granular size below 10mm for at least 90% of cargo; maximum of 40% below 150 micron			Size below 6.3mm max 15% Size above 31.15mm max 25%
Pricing Point	Qingdao Port (FOT and CFR respectively)			FOT Qingdao Port
Timing (Seaborne)	Loading within 4 weeks, Delivery within 8 weeks			Delivery within 2 weeks
Payment Terms	L/C at sight			L/C at sight or CAD

Iron Ore Index Compilation Rationale and Data Exclusions

MMi iron ore indices are compiled from data provided by companies that are part of the iron ore supply chain and involved in spot market transactions. The indices are calculated using detailed transaction-level data submitted to MMi by these companies. This data is normalised to the appropriate specifications and screened to remove outliers before volume-weighted average prices are calculated from the remaining core set of data.

For more details on MMi's iron ore methodology please download the guide published on our website at: www.mmiprices.com

Data Exclusions*

	62%	58%	65%
Port Index	0	0	0
Seaborne index	0		0
Lump Index 62.5	0		

* Number of price submissions for iron ore indices that were excluded from index calculations today as they were anomalous and could not be verified

IRON ORE DOMESTIC CONCENTRATE INDEX CALCULATION METHODOLOGY

The compilation method for price index generally refers to the compilation method of CPI price index and other price indices, breakdown the price data and calculate the average value according to a certain method, taking the vertical axis as the regional composite index (average of different grade index) and the horizontal axis as the grade composite index (average of different regional index), a total composite index for domestic ore can be output ultimately. The process system is also adopted in the calculation i.e. each sub-index can be obtained as well.

AVERAGE IRON ORE SPECIFICATIONS APPLIED FOR BRAND PRICE ASSESSMENTS

PORT STOCK BRANDS

Mar 1st, 2023	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.88%	2.35%	4.45%	0.055%	9.26%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines	61.64%	2.31%	3.87%	0.099%	9.45%
Newman Fines	62.49%	2.31%	3.98%	0.090%	7.72%
MAC Fines	60.43%	2.38%	4.64%	0.078%	7.57%
Jimblebar Blended Fines	60.87%	2.95%	4.53%	0.111%	7.32%
Carajas Fines	64.90%	1.39%	1.82%	0.080%	7.78%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	62.56%	1.58%	4.79%	0.091%	8.72%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

Mar 1st, 2023	Specifications applied for 58% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
SSF	56.49%	3.20%	6.19%	0.065%	9.18%
FMG Blended Fines	58.20%	2.43%	5.54%	0.057%	8.29%
Robe River	56.44%	3.16%	5.73%	0.042%	8.44%
Western Fines	57.88%	2.87%	7.50%	0.062%	7.45%
Atlas Fines	58.00%	1.85%	5.50%	0.090%	9.00%
Yandi	56.87%	1.58%	6.41%	0.042%	9.53%

SEABORNE BRANDS

	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.70%	2.30%	4.90%	0.055%	8.00%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines 62%	62.00%	2.60%	4.30%	0.090%	9.50%
Newman Fines	62.30%	2.40%	4.30%	0.095%	8.00%
MAC Fines	60.80%	2.35%	4.40%	0.075%	7.80%
Jimblebar Blended Fines	60.50%	3.10%	4.80%	0.120%	7.80%
Carajas Fines	65.10%	1.50%	1.70%	0.080%	8.50%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	63.00%	1.50%	5.00%	0.070%	7.00%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

BLOOMBERG TICKERS

PORT STOCK INDICES

	FOT Qingdao (RMB/wet tonne)	CFR Qingdao Equivalent (USD/dry tonne)
IOPI62	IRCNQ001	IRCNQ004
IOPI58	IRCNQ002	IRCNQ005
IOPI65	IRCNQ003	IRCNQ006
IOPLI62	IRCN0036	IRCN0037

SEABORNE INDICES

	CFR Qingdao (USD/DMT)
IOSI62	IRCN0034
IOSI65	IRCN0035

CONTACT US

MMI Singapore Office Details: Level 28, Manulife Tower, 8 Cross Street Singapore. Tel: + 65 6850 7629 E: prices@mmiprices.com
SMM Singapore Office Details: Level 28, Manulife Tower, 8 Cross Street Singapore. Tel: + 65 6850 7630 E: service.en@smm.cn
SMM Shanghai Office Details: 9th FL, Building 9, Lujiazui Software Park, No.20, Lane 91, Pudong, Shanghai Tel: +86 021 5155 0306 E: service.en@smm.cn

Visit www.mmiprices.com for full index price histories and archive of daily reports

This information has been prepared by Metals Market Index ("MMi"). Use of the information presented here is at your sole risk, and any content, material and/or data presented or otherwise obtained through your use of the information in this document is at your own discretion and risk and you will be solely responsible for any damage to you personally or your company or organisation or business associates whatsoever which in anyway results from the use, reliance or application of such content material and/or information. Certain data has been obtained from various sources and any copyright existing in such data shall remain the property of the source. Except for the foregoing, MMi retains all copyright within this document. The copying or redistribution of any part of this document without the express written authority of MMi is forbidden.

COPYRIGHT METALS MARKET INDEX, ALL RIGHTS RESERVED